

Dennis Delali Kwesi Wayo

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Summary

Research engineer building fault-tolerant quantum software for photonic and hybrid quantum architectures. Experience spans decoder benchmarking, architecture-level simulation, and reproducible scientific-computing pipelines in Python, Rust, and C++. Built end-to-end QEC tooling across hardware-to-decoder interfaces, with emphasis on decoder comparability, reproducibility, and architecture-level performance analysis. Current work sits at the boundary between scalable simulation, decoder infrastructure, and compiler-aware hardware modeling for fault-tolerant quantum systems.

Core Skills

Programming: Python, Rust, C++, Swift, Julia, Bash

Quantum Software: Calibration and benchmarking workflows, hardware-control interfaces, quantum error correction, surface-code threshold workflows, decoder benchmarking (BP/MWPM/UF), GKP digitization, photonic architecture modeling

Quantum Tooling: PennyLane, Qiskit, Qibo, Qibolab, Qibocal, Mitq, QuTiP

Scientific Computing: Numerical linear algebra, HPC Linux workflows, parallel simulation, data pipelines, reproducibility and artifact tracking

Software Engineering: Git/GitHub, code review, CI/CD (GitHub Actions), testing (pytest), documentation, packaging, Docker, REST-style API development

ML for Science: PyTorch, TensorFlow, JAX, differentiable simulation

Quantum Software Engineering Experience

Gottesman Software — Public Quantum Research Software Portfolio 2026–Present

- Founded and maintain a public GitHub organization for open-source quantum research software: github.com/Gottesman-Software.
- Organize QEC, photonic simulation, and driver-layer prototypes including LiDMaS+, SchroSIM, and photon-qdrivers; public site: gottesman-software.github.io.
- Use the organization as a visible home for reproducible research software, project documentation, issue tracking, and reviewable development artifacts.

LiDMaS+ — Logical Decoder Benchmarking Engine (C++/Rust/Python) 2026–Present

- Built replay-driven QEC benchmarking pipelines with normalized decoder I/O to compare BP, MWPM, UF, and neural-style adapters on identical syndrome streams.
- Added diagnostics for syndrome satisfaction, residual nonzero rate, flip sparsity, and decoder stability, supporting transparent decoder comparisons across matched datasets.
- Implemented deterministic reproducibility checks, artifact hashing, provider onboarding APIs, and run tracking for repeatable QEC experiments.

SchroSIM — Photonic/CV Quantum Simulation Stack (Swift/Rust/Python) 2024–Present

- Built a layered photonic-circuit platform separating circuit authoring, compilation, numerical execution, and hardware mapping for architecture-level simulation.
- Implemented Gaussian simulation pathways and selected non-Gaussian routines for continuous-variable and GKP-motivated studies.

- Developed interfaces for hardware-aware CV simulation under loss, dephasing, and detector modeling assumptions.

Research Assistant — Nazarbayev University

2021–Present

- Developed multiphysics and quantum-adjacent simulation workflows with finite-difference methods, HPC execution pipelines, and reproducible experiment structure.
- Built DFT/TDDFT simulation stacks for photonic material modeling and spectral analysis.
- Integrated ML-assisted scientific workflows with physics-based simulation and documented data handling.

Selected Open-Source Quantum Software Contributions

- **Qiskit** — Extended `QuantumCircuit` standard and controlled-gate helpers to propagate gate labels, with regression tests #16373; refactored QPY roundtrip load-test setup #16372.
- **PennyLane** — Enabled documentation testing for `pennylane.shadows`, corrected executable classical-shadow examples, and added deterministic validation; merged PR #9566.
- **PennyLane** — Improved `qml.assert.equal` diagnostics for measurement-process mismatches with regression tests and changelog entry; open PR #9605.
- **PennyLane** — Added text-drawer multiplexer rendering for `SelectPauliRot` #9604 and improved `ExecutionConfig` print readability #9603.
- **Qibolab** — Added a Dynamiqs simulation engine for emulator and hardware-control development workflows, with dependency integration and focused emulator tests; merged PR #1479.
- **Qibolab** — Fixed emulator evolution and result-handling paths after storage/GPU changes, including QuTiP/Dynamiqs compatibility and cyclic probability marginalization validated with Qibocal Rabi smoke runs; merged PRs #1497 and #1505.
- **Qibocal** — Fixed emulator example-platform drive configuration for current Qibolab schemas and validated Qibocal Rabi smoke runs on qubit/qutrit platforms; merged PR #1551.
- **Mitiq/Qibo** — Developed a Qibo tutorial combining repetition-code syndrome checks, post-selection/correction, noise sweeps, and zero-noise extrapolation for reproducible benchmarking workflows; open PR #3023.

Technical Leadership, Documentation & Community

- Develop maintainer-ready open-source contributions with focused tests, issue context, documentation updates, and PR narratives suitable for review by distributed quantum software teams.
- Prepare developer-facing examples and tutorials that translate quantum algorithms, error mitigation, calibration-adjacent workflows, and benchmarking concepts into executable software artifacts.
- Qiskit Global Summer School Development Week participant and scheduled QGSS mentor (2026); IBM Qiskit Advocate and QAMP Mentor (2025).

Education

PhD Candidate, Computer Science (extended research mode)

Expected 2029

TU Bergakademie Freiberg, Faculty of Mathematics and Computer Science, Germany

Mode: Research-based and flexible; no taught-course classroom attendance requirement

Research focus: Parallel knowledge-graph and AI-augmented compiler architectures for scalable logical-GKP quantum error correction in continuous-variable quantum systems

M.S. Candidate, Computer Science (OMSCS)

Expected 2027

Georgia Institute of Technology, College of Computing, Atlanta, USA

Specialization: Computing Systems; relevant preparation in algorithms, quantum computing/hardware, machine learning, NLP, HPC, high-performance computer architecture, software architecture & design, embedded systems optimization, and advanced operating systems

PhD, Chemical Engineering	2026
Universiti Malaysia Pahang, FTKKP, Malaysia	
<i>Research focus:</i> Quantum photonic-induced solid-state electrolyte for CO ₂ reduction	
M.Sc. Petroleum Engineering	2022
Nazarbayev University, Kazakhstan	
B.Sc. Petroleum Engineering	2019
Kwame Nkrumah University of Science and Technology, Ghana	
A.S. in Mechanical Engineering	2014
Tamale Technical University, Ghana	

Selected Publications & Preprints

- Wayo, D.D.K. et al. (2026). Hardware-in-the-loop syndrome-to-decoder validation for repetition, surface, CSS-LDPC, and digitized-GKP codes. *arXiv preprint* 2607.19447.
- Wayo, D.D.K. et al. (2026). Decoder comparability across quantum software stacks: repeated-round surface and digitized-GKP syndrome replay. *arXiv preprint* 2607.19446.
- Wayo, D.D.K. et al. (2026). A unified hardware-to-decoder architecture for hybrid continuous-variable and discrete-variable quantum error correction in LiDMaS+. *arXiv preprint* 2604.15389.
- Wayo, D.D.K. et al. (2026). Decoder dependence in surface-code threshold estimation with native Gottesman–Kitaev–Preskill digitization and parallelized sampling. *arXiv preprint* 2603.25757.
- Wayo, D.D.K. et al. (2026). Decoder dependence in surface-code threshold estimation under digitized hybrid continuous-variable and discrete noise. *Fortschritte der Physik*, 74(6). DOI: 10.1002/prop.70124.
- Wayo, D.D.K. & Groppe, S. Quantum Machine Learning Codebook. *SpringerBriefs in Computer Science*. Springer.
- Wayo, D.D.K. & Groppe, S. (2026). RaCS: near-zero-error classical data encoding on photonic quantum processors via redundancy-assisted coherent-state codes. *Fortschritte der Physik*, 74(4), e70095. DOI: 10.1002/prop.70095.
- Wayo, D.D.K. et al. (2026). DifGa: Differentiable error mitigation for multi-mode gaussian and non-gaussian noise in quantum photonic circuits. *Journal of Physics: Photonics*. DOI: 10.1088/2515-7647/ae7864.
- Wayo, D.D.K. (2026). Simulation of ultrafast photonic circuits via nonlinear Schrödinger dynamics and quantum detector modeling. *Optical and Quantum Electronics*, 58(3). DOI: 10.1007/s11082-026-08700-y.
- Wayo, D.D.K. et al. (2025). Atomistic modeling of rare earth ions in photonic materials. *Luminescence*, 40(9). DOI: 10.1002/bio.70297.
- Wayo, D.D.K. (2025). Ensembles of graph and physics-informed machine learning for scientific modeling in materials science: A review. *Archives of Computational Methods in Engineering*, 33(1), 963–988. DOI: 10.1007/s11831-025-10325-5.

Honors, Mentoring & Service

- IBM Qiskit Advocate and QAMP Mentor (2025); Qiskit Global Summer School Development Week participant and scheduled QGSS mentor (2026).
- Womanium & WISER Quantum Scholar (2025).
- Reviewer: Journal of Open Source Software, Scientific Reports, Nanoscale Advances.